MA'ADEN معادن

Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project Customer: **Mansourah Massarah Gold Project**
MA'ADEN GOLD AND BASE METALS (MA'ADEN)

Title: **GENERAL ARRANGEMENT DRAWING**
POWER PLANT STORAGE TANK AREA
G.A. AND DETAIL OF GRADE SLAB

Document Number: **MD-216-49PR-EG-CV-GAD-2009**
Revision: **D01**
Revision Date: **03.05.2023**
Status: **ISSUED FOR CONSTRUCTION**

Outotec Document Number:	900707	GAU01	WG01	MTL01	45081	OU100688139	6
L&T Document Number:	O19045	008	C	G	24292	005	6

REVISION HISTORY

Rev	Date	Revision Text	Prepared By	Reviewed By	Approved By	Employer
A01	16.09.2020	ISSUED FOR REFERENCE	AMJ	SGC	PRI	
C01	16.10.2020	ISSUED FOR CONSTRUCTION & REVISED AS MKD. C01	AMJ	SGC	PRI	
C02	06.02.2021	ISSUED FOR CONSTRUCTION & REVISED AS MKD. C02	PDK	SGC	PRI	
C03	12.04.2021	ISSUED FOR CONSTRUCTION & REVISED AS MKD. C03	AMJ	SGC	PRI	
C04	07.07.2021	ISSUED FOR CONSTRUCTION & REVISED AS MKD. C04	SYA	SJM	AG	
C05	24.08.2021	ISSUED FOR CONSTRUCTION & REVISED AS MKD. C05	PML	SJM	AG	
D01	03.05.2023	AS BUILT	DSR	SJM	AG	

DRAWING NUMBER: MD-216-49PR-EG-CV-GAD-2009 D01

LAYOUT OF GRADE SLAB FOR STORAGE TANK AREA

SECTION Z - Z (SCALE 1:50)

TYP. DETAIL OF PIT MKD. PT12 (SCALE 1:50)

TYP. DETAIL OF PIT MKD. PT8A & PT8B (SCALE 1:50)

SECTION 5' - 5' (SCALE 1:50)

SECTION 5' - 5' (SHOWING RC DETAIL) (SCALE 1:50)

DETAIL - X (SCALE 1:20)

SECTION Y - Y (SCALE 1:25)

TYP. DETAIL OF PIT MKD. PT11

SECTION Z - Z (SCALE 1:10)

SECTION 3 - 3 (SCALE 1:40)

TYP. DETAIL OF PIT MKD. PT4 (SCALE 1:40)

TYP. DETAIL OF PIT MKD. PT5, PT6, PT7 & PT9 (SCALE 1:50)

TYP. DETAIL OF PIT MKD. PT2

TYP. DETAIL OF PIT MKD. PT3 (SCALE 1:50)

TYP. DETAIL OF PIT MKD. PT1 (SCALE 1:40)

SECTION 1-1 (SCALE 1:40)

SECTION 2-2 (SCALE 1:40)

DETAIL - Y (SCALE 1:10)

DETAIL - Z (SCALE 1:10)

SECTION 4 - 4 (SCALE 1:50)

TYP. DETAIL OF PIT MKD. PT8 (SCALE 1:50)

SECTION 5 - 5 (SCALE 1:40)

TYP. DETAIL OF PEDESTAL MKD. P1A, P1B & P1C (SCALE 1:40)

TYP. DETAIL OF PEDESTAL MKD. P3 & P5 (SCALE 1:40)

SECTION : H - H (SCALE - 1:40)

DETAIL OF PEDESTAL MKD. P4 (ORIENTATION AS PER LAYOUT PLAN) (SCALE - 1:40)

TYP. DETAIL OF PEDESTAL MKD. P1 (SCALE 1:25)

SECTION A - A (SCALE 1:25)

SECTION E - E (SCALE 1:25)

SECTION F - F (SCALE 1:20)

TYP. DETAIL OF PEDESTAL MKD. P2 (ORIENTATION AS PER LAYOUT PLAN) (SCALE 1:20)

TYP. DETAIL OF PIT MKD. PT10 (SCALE 1:50)

SECTION G - G (SCALE 1:40)

SECTION S1 - S1 (SCALE 1:10)

DETAIL OF INSERT PLATE - IP1 (SCALE 1:10)

SECTION B - B (SCALE 1:50)

SECTION C - C (SCALE 1:50)

SECTION : K - K (SCALE - 1:40)

DETAIL OF PEDESTAL MKD. P2A (SCALE - 1:40)

SECTION J - J (SCALE 1:40)

TYP. DETAIL OF PEDESTAL MKD. P6 (SCALE 1:40)

SECTION : 1A - 1A (SCALE - 1:40)

SECTION 1A - 1A (SHOWING R.C. DETAIL ONLY) (SCALE 1:40)

SECTION Y - Y (SHOWING RC DETAIL) (SCALE 1:30)

SECTION D - D (SCALE 1:50)

SECTION A - A (SHOWING RC DETAIL) (SCALE 1:25)

SECTION B - B (SHOWING RC DETAIL) (SCALE 1:50)

SECTION E - E (FOR PEDESTAL MKD. P1A, P1B & P1C) (SHOWING RC DETAIL) (SCALE 1:25)

SECTION F - F (SHOWING RC DETAIL) (SCALE 1:25)

SECTION : J - J (SHOWING R.C. DETAIL ONLY) (SCALE - 1:25)

SECTION 1-1 (SHOWING RC DETAIL) (SCALE 1:40)

TYP. RC DETAIL OF 150 THK. LIGHT DUTY GRADE SLAB (SCALE 1:25)

SECTION : K1 - K1 (SCALE - 1:40)

SECTION A1 - A1 (SCALE 1:25)

SECTION C - C (SHOWING RC DETAIL) (SCALE 1:50)

SECTION G - G (SHOWING R.C. DETAIL ONLY) (SCALE 1:25)

SECTION H1 - H1 (SHOWING RC DETAIL) (SCALE 1:25)

TYP. R.C. SECTION OF FOUNDATION MKD. P2A (SCALE - 1:40)

SECTION Z1 - Z1 (SCALE 1:50)

TYP. DETAIL OF BEAM MKD. B1 (SCALE 1:20)

SECTION 3 - 3 & 4 - 4 (SHOWING RC DETAIL) (SCALE 1:50)

SECTION 2-2 (SHOWING RC DETAIL) (GRADE SLAB REIN. NOT SHOWN HERE FOR CLARITY) (SCALE 1:40)

SECTION 5 - 5 (SHOWING RC DETAIL) (SCALE 1:50)

KEY PLAN (SCALE 1:1000)

LEGEND :-

RL - REDUCED LEVEL.
CL - CENTRE LINE.
T.O.C - TOP OF CONCRETE.
T.O.G - TOP OF GROUT
F.G.L - FINISHED GROUND LEVEL.
P.C.D - PITCH CIRCLE DIAMETER.
TYP. - TYPICAL.
PCC - PLAIN CEMENT CONCRETE.
MKD - MARKED.
Ø - DIAMETER.
R - RADIUS.
I.L - INVERT LEVEL.
F.F.L - FINISHED FLOOR LEVEL
I.D - INNER DIAMETER.
O.D - OUTER DIAMETER.
THK. - THICK.
DET. - DETAIL.
TNK. - TANK.
FDN. - FOUNDATION.
OS - OIL WATER DRAINAGE PIPE
RS - STORM WATER DRAINAGE PIPE

NOTES :-

1. ALL DIMENSIONS ARE IN MILLI METRES AND LEVELS AND CO-ORDINATES ARE IN METRES. UNLESS OTHERWISE STATED.
2. GRADE OF CONCRETE SHALL BE 35 MPa FOR FOUNDATION & 15 MPa FOR BLINDING CONCRETE.
3. GRADE OF REINFORCEMENT SHALL BE DEFORMED BAR 420 MPa (GRADE 60) - ASTM A615/A615M.
4. NET SAFE BEARING CAPACITY OF SOIL AT BOTTOM OF FOUNDATION CONSIDER 500KN/ SQ. M.
5. CLEAR COVER TO MAIN REINFORCEMENT SHALL BE FOUNDATIONS = 50 MM , WALLS = 50 MM , GRADE SLAB = 25 MM , PEDESTAL = 50 mm.
6. IF ANY PART OF FOUNDATION COMES ON BACKFILL ZONE OF ADJACENT FOUNDATION OR ANY LOOSE SOIL ARE ENCOUNTERED BELOW FOUNDATION LEVEL THE SAME SHALL BE REMOVED AND FILLED WITH PCC.
7. ALL OTHER NOTES REFER DRAWING NO. MD-101-SMEM-EG-ST-DST-0001 TO 0008.
8. GENERAL NOTES FOR CIVIL WORKS & STANDARD REINFORCEMENT DETAIL REFER DRAWING NO. MD-216-1000-EG-CV-DED-1001 & 1005.
9. FOR ANCHOR BOLTS STANDARD DETAILS. REFER DRAWING NO. MD-216-1000-EG-CV-DED-1003
10. THIS DRAWING SHALL BE READ IN CONJUNCTION WITH DRAWING NO. MD-216-49PR-EG-ST-GAD-2413.
11. PIPING MATERIAL STANDARD - AT12.
12. ALL LEVELS AS MENTIONED ARE BOTTOM OF PIPE.
13. OILY WATER DRAINAGE TO BE ADJUSTED ACCORDING TO SITE CONDITION.
14. ALL PITS MARKED WITH VALVE SYMBOL SHALL BE PROVIDED WITH 100 MM EXTENDED AT12 PIPE LINE INSIDE PIT.
15. FOR VALVE FIXING DETAILS INSIDE VALVE PIT SEPARATE DRAWING WILL BE PROVIDED.
16. PIPE SHALL BE EXTENDED BY 100 MM INSIDE FOR THESE PITS TO INSTALL VALVES AT A LATER STAGE.
17. PLEASE NOTE ALL PIT PIPE CONNECTION SHALL BE DONE AS PER THE PLAN DRAWING. IN ELEVATION OF THE PITS THE PIPE CONNECTION ARE NOT SHOWN.

SHEET 1 OF 2

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		D01	03/05/2023	AS-BUILT AND REVISED AS MKD.	PML	SJM	AVG	AG	
DBAF507559	STORAGE TANK AREA - INFORMATION DRAWING	C05	24/08/2021	ISSUED FOR CONSTRUCTION AND REVISED AS MKD.	PML	SJM	AVG	AG	
DBAF523403	LOAD INPUT - STORAGE TANK AREA	C04	07/07/2021	ISSUED FOR CONSTRUCTION AND REVISED AS MKD.	SYA	SJM	AVG	AG	
MD-216-49PR-EG-ME-LAD-V553-1002	LAYOUT OF POWER GENERATION	C03	12/04/2021	ISSUED FOR CONSTRUCTION AND REVISED AS MKD.	AMJ	SGC	SHD	PRI	
MD-216-1000-EG-ST-CRT-1001	CIVIL, STRUCTURAL & ARCHITECTURAL - DESIGN CRITERIA	C02	06/02/2021	ISSUED FOR CONSTRUCTION AND REVISED AS MKD.	PDK	SGC	SHD	PRI	
MD-216-49PR-EG-ME-GAD-V553-1042	LFO STORAGE TANK	C01	19/10/2020	ISSUED FOR CONSTRUCTION AND REVISED AS MKD.	AMJ	SGC	SHD	PRI	
MD-216-49PR-EG-ME-GAD-V553-1041	HFO STORAGE TANK	A01	16/09/2020	ISSUED FOR REFERENCE	AMJ	SGC	SHD	PRI	

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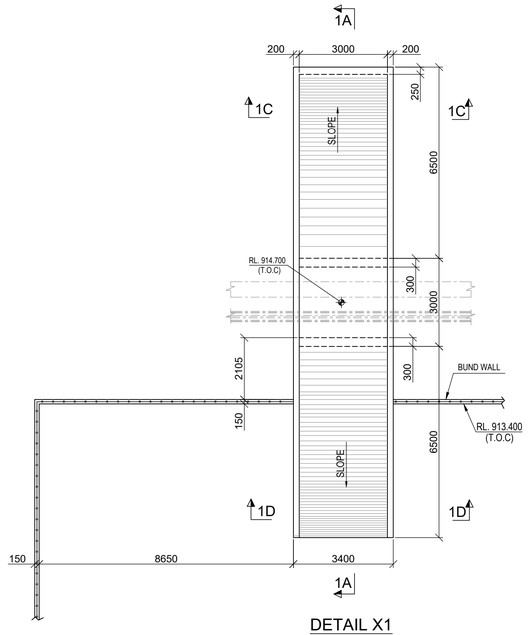
LARSEN & TOUBRO DRAWING NUMBER	REVISION
019045_008_C_G_24292_005	6
OUTOTEC DRAWING NUMBER	DOCUMENT ID / REVISION
900707 __ GAU01 __ WG01 __ MTL01 __ 45081	OU100688139 / 6

DRAWN	AMJ	05/09/2020
CHECKED	SGC	08/09/2020
ENGINEER	SHD	05/09/2020
APPROVED	PRI	16/09/2020
EMPLOYER		
SCALE	1:100, 1:40, 1:20	A0

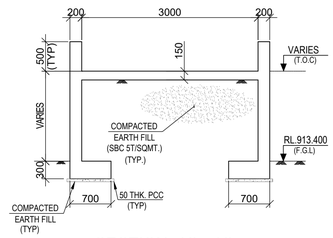
MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT
POWER PLANT STORAGE TANK AREA
GA AND DETAIL OF GRADE SLAB

DRAWING NUMBER: MD - 216 - 49PR - EG - CV - GAD - 2009 | REV: D01

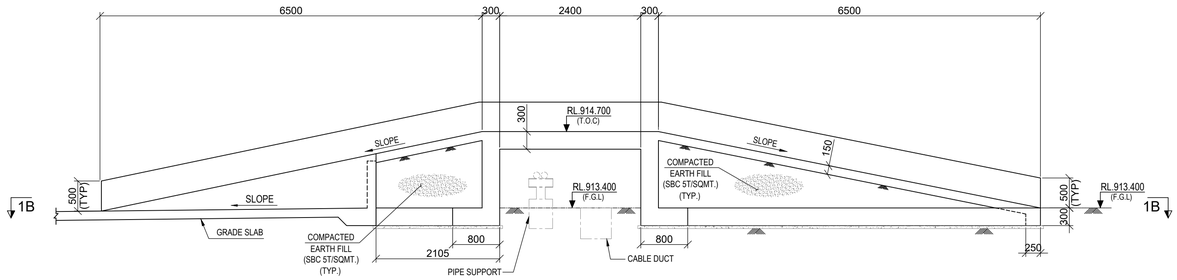
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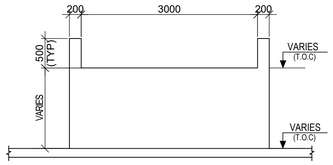
DETAIL X1
(SCALE 1:50)



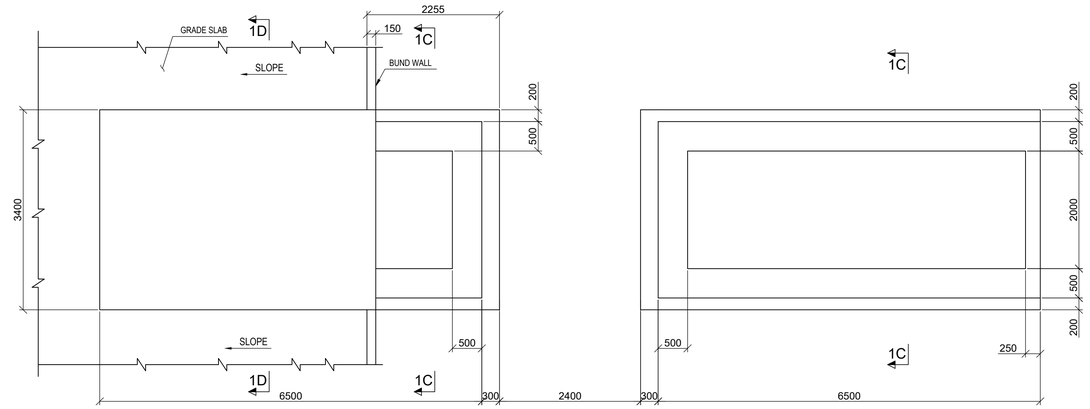
SECTION 1C - 1C
(SCALE 1:50)



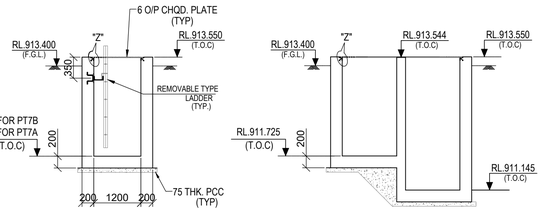
SECTION 1A - 1A
(SCALE 1:50)



SECTION 1D - 1D
(SCALE 1:50)

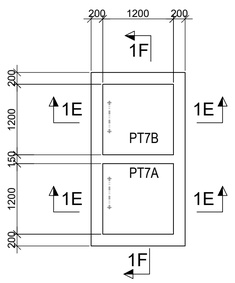


SECTION 1B - 1B
(SCALE 1:50)

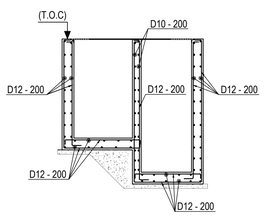


SECTION 1E - 1E
(SCALE 1:50)

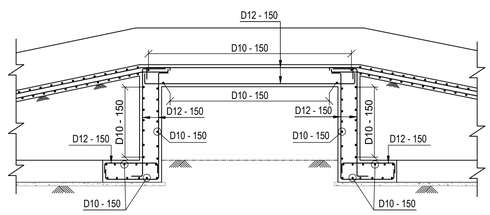
SECTION 1F - 1F
(SCALE 1:50)



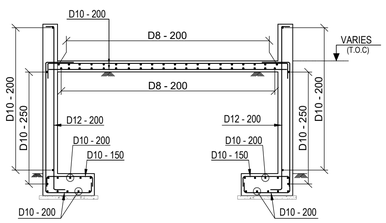
TYP. DETAIL OF PIT MKD. PT7A, PT7B
(SCALE 1:50)



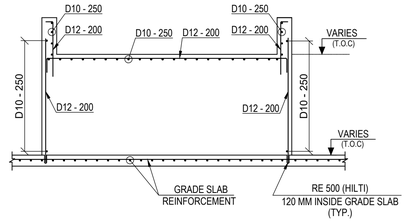
SECTION 1F - 1F
(SHOWING RC DETAIL)
(SCALE 1:50)



SECTION 1A - 1A
(SHOWING RC DETAIL)
(SCALE 1:40)



SECTION 1C - 1C
(SHOWING RC DETAIL)
(SCALE 1:40)



SECTION 1D - 1D
(SHOWING RC DETAIL)
(SCALE 1:40)

LEGEND :-

RL. - REDUCED LEVEL.
CL. - CENTRE LINE.
T.O.C. - TOP OF CONCRETE.
T.O.G. - TOP OF GROUT
F.G.L. - FINISHED GROUND LEVEL.
P.C.D. - PITCH CIRCLE DIAMETER.
TYP. - TYPICAL.
PCC. - PLAIN CEMENT CONCRETE.
MKD. - MARKED.
Ø - DIAMETER.
R - RADIUS.
I.L - INVERT LEVEL
F.F.L - FINISHED FLOOR LEVEL
I.D - INNER DIAMETER.
O.D - OUTER DIAMETER.
THK. - THICK.
DET. - DETAIL.
TNK. - TANK.
FDN. - FOUNDATION.
OS. - OILY WATER DRAINAGE PIPE
RS. - STORM WATER DRAINAGE PIPE

NOTES :-

1. ALL DIMENSIONS ARE IN MILLI METRES AND LEVELS AND CO-ORDINATES ARE IN METRES. UNLESS OTHERWISE STATED.

SHEET 2 OF 2

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		D01	03/05/2023	AS-BUILT	PML	SJM	AVG	AG	
DBAF507559	STORAGE TANK AREA - INFORMATION DRAWING	C05	24/08/2021	ISSUED FOR CONSTRUCTION AND REVISED AS MKD. 05	PML	SJM	AVG	AG	
DBAF523403	LAYOUT OF POWER GENERATION	C04	07/07/2021	ISSUED FOR CONSTRUCTION AND REVISED AS MKD. 04	SYA	SJM	AVG	AG	
MD-216-49PR-EG-ME-LAD-V553-1002	LAYOUT OF POWER GENERATION	C03	12/04/2021	ISSUED FOR CONSTRUCTION AND REVISED AS MKD. 03	AMJ	SGC	SHD	PRI	
MD-216-1000-EG-ST-CRT-1001	CIVIL, STRUCTURAL & ARCHITECTURAL - DESIGN CRITERIA	C02	06/02/2021	ISSUED FOR CONSTRUCTION AND REVISED AS MKD. 02	PDK	SGC	SHD	PRI	
MD-216-49PR-EG-ME-GAD-V553-1042	LFO STORAGE TANK	C01	19/10/2020	ISSUED FOR CONSTRUCTION AND REVISED AS MKD. 01	AMJ	SGC	SHD	PRI	
MD-216-49PR-EG-ME-GAD-V553-1041	HFO STORAGE TANK	A01	16/09/2020	ISSUED FOR REFERENCE	AMJ	SGC	SHD	PRI	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

LARSEN & TOUBRO DRAWING NUMBER: O19045_008_C_G_24292_005 — REVISION: 6
OUTOTEC DRAWING NUMBER: 900707__GAU01__WG01_MTL01__45081 — DOCUMENT ID: OU100688139 — REVISION: 6

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DRAWN	PML	05/09/2020	MANSOURAH - MASSARAH GOLD PROJECT
CHECKED	SJM	08/09/2020	GENERAL ARRANGEMENT
ENGINEER	AVG	05/09/2020	POWER PLANT STORAGE TANK AREA
APPROVED	AG	16/09/2020	GA AND DETAIL OF GRADE SLAB AND RAMP
EMPLOYER			
SCALE 1:30		A1	DRAWING NUMBER: MD-216-49PR-EG-CV-GAD-2009 — REV: D01